SERVICE MANUAL

tCOM+ Transcutaneous Monitor & Sensors



sentec.

Warranty

The manufacturer warrants to the initial purchaser that each new tCOM+ will be free from defects in workmanship and materials. The manufacturer's sole obligation under this warranty is to at its own choice repair or replace any monitor – for which the manufacturer acknowledges the warranty cover – with a replacement monitor.

Warranty Exclusions and System Performance

Sentec can neither guarantee nor verify instrument performance characteristics nor accept warranty claims or product liability claims if the recommended procedures are not carried out, if the product has been subject to misuse, neglect or accident, if the product has been damaged by extraneous causes, if accessories other than those recommended by Sentec are used, if the warranty seal on the lower side of the monitor is broken, or if instrument repairs are not carried out by Sentec authorized service personnel.

Unauthorized modifications to Sentec products could void your warranty and alter the regulatory status of the devices. Any resulting service required is not covered under our service agreements. Such modifications can affect the performance or safety of your device in unpredictable ways, and Sentec is not responsible for equipment that has been modified.

CAUTION: Federal law (U.S.) restricts this device to sale by or on the order of a physician.

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Abbreviations

AHP	Absolute Heating Power
CO2	Carbon dioxide
DS	Docking Station (calibration unit integrated in the tCOM+)
HP	Heating Power
HPM	Heating Power Mode
LED	Light emitting diode
MRI	Magnetic Resonance Imaging
O2	Oxygen
PaCO2	Arterial carbon dioxide partial pressure
PaO2	Arterial oxygen partial pressure
PcCO2	Cutaneous carbon dioxide partial pressure (i.e. the CO2 partial pressure at the skin surface)
PCO2	Used to display/label tcPCO2 on the tCOM+ and – unless explicitely stated otherwise – throughout this manual
PcO2	Cutaneous oxygen partial pressure (i.e. the O2 partial pressure at the skin surface)
PI	Pulsation Index
PO2	Used to display/label tcPO2 on the tCOM+ and – unless explicitely stated otherwise - throughout this manual
POST	Power-On Self-Test
RO	Responsible Organization
PR	Pulse rate

RHP	Relative Heating Power
RMI	Remote monitoring interrupted
SaO2	Arterial oxygen saturation
SDMS	Sentec Digital Monitoring System
SpO2	Functional oxygen saturation of arterial hemoglobin as measured with a pulse oximeter
TC	Transcutaneous
tCOM+	Sentec Patient Monitor (Transcutaneous Carbon Dioxide and Oxygen Monitor)
tcPCO2	Transcutaneous carbon dioxide partial pressure, i.e., an estimate of PaCO2 calculated from the measured PcCO2 and displayed/labeled on the tCOM+ and - unless explicitly stated otherwise - throughout this manual as 'PCO2'
tcPO2	Transcutaneous oxygen partial pressure, i.e., an estimate of PaO2 calculated from the measured PcO2 and displayed/labeled on the tCOM+ and - unless explicitly stated otherwise - throughout this manual as 'PO2'

1 Introduction

This Manual provides information on

- **maintenance and safety and functionality tests** for the Sentec Digital Monitoring System (using tCOM+) intended for **qualified personnel*** (*see section 3*)
- **troubleshooting** for the SDMS (*see section 4*). The troubleshooting list covers two levels of recommended troubleshooting procedures a) for the **operator****, b) for **qualified personnel***
- **Replacement of service parts** intended for **qualified personnel* only** (*see section 5*).

*(** Operator is every routined user of the SDMS, e.g. nurse.)*
(Qualified personnel should have obtained a relevant technical education/degree, e.g., medical or maintenance technicians.)*

Note: Repairs that require opening the tCOM+ cover must be done by Sentec only.

This manual must be read carefully prior performing service. Section 1.2 *"Safety Information"* must be read and understood.

1.1 Related Documents

To perform service, you must know how to operate the SDMS. The SDMS consists of the tCOM+ monitor, sensors, and accessories.

Refer to the following Manuals and Directions for Use:

Table 1 Related Documents

System component	Manual	Content
SDMS using tCOM+	HBQ-179 Instruction Manual for the tCOM+ Transcutaneous Monitor & Sensors	Getting started / Using the SDMS: The *Instruction Manual for the tCOM+ and the Sentec Transcutaneous Sensors* highlights the essential points when using the SDMS. To ensure proper operation of the SDMS, precisely follow the instructions provided in this manual step by step. Contains routine checks and maintenance procedures that can be performed by the operator.
SDMS using tCOM+	HBQ-197 Service Manual for the tCOM+ Transcutaneous Monitor & Sensors (present manual)	Provides information on maintenance, safety, and functionality tests and troubleshooting procedures as well as on service/maintenance procedures for sensors.
SDMS using tCOM+	HBQ-122 Cleaning and Disinfection	List of tested cleaning and disinfection solutions suitable for Sentec products.
Sensors / Disposables	Respective Directions for Use	Additional information on Sentec TC Sensors, the Membrane Changer and Membrane Changer Insert, the Ear Clip, the Multi-Site Attachment Rings, and Sentec's Staysite™ Adhesive is provided in the respective Directions for Use.

Note: Related forms (as referenced within this manual) are listed in section 8 "Appendix".

Note: Information / Instructions about the Sensor Adapter Cable, the Contact Gel and the Calibration Gas are provided in the Instruction Manual for the tCOM+ Transcutaneous Monitor & Sensors.

Note: SDMS related tutorials are available for online viewing at www.sentec.com/tv

1.2 Safety Information

Note: The Sentec Digital Monitoring System (SDMS) is to be operated by qualified personnel only. Read this manual, accessory Directions for Use, all precautionary information, and specifications before use.

Warnings

This symbol identifies a **WARNING** statement. A warning alerts the reader about a situation which, if not avoided, could result in death or serious injury to the patient, user, or serious adverse events on the environment respectively.



Cautions

This symbol identifies a **CAUTION** statement. The term caution is used for the statement of a hazard alert that warns the reader of a potentially hazardous situation which, if not avoided, may result in minor or moderate injury to the user or patient or damage to the equipment or other property. It may also be used to alert against unsafe practices. This includes the special care necessary for the safe and effective use of the device and the care necessary to avoid damage to a device that may occur as a result of use or misuse.



1.2.1 General Safety Information

⚠ **WARNING:** Do not use tCOM+, sensors, cables, or connectors that appear damaged.

⚠ **WARNING:** Carefully route and fix cables to reduce the possibility of patient entanglement or strangulation.

⚠ **WARNING:** To ensure patient safety, do not place the tCOM+ in any position that might cause it to fall on the patient.

⚠ **WARNING:** Keep the tCOM+ (as well as any discarded parts) out of reach of children under the age of 5 years. Some parts of the tCOM+ are small enough to be swallowed and may block the trachea.

⚠ **WARNING:** Explosion and flammability hazards. Do not use the tCOM+ in the presence of flammable anesthetics / gases or other flammable substances in any environment which has increased oxygen content.

⚠ **WARNING:** To ensure protection of patient, operator, and equipment from the effects of defibrillation and diathermy/ electro-surgery, cables manufactured by Sentec must be used.

⚠ **WARNING:** The tCOM+ should not be used adjacent to or stacked with other equipment as these can cause electromagnetic interference and thereby result in incorrect measurements. If adjacent or stacked use is necessary, the tCOM+ should be observed to verify normal operation in the configuration it is to be used.

⚠ **WARNING:** The use of accessories, sensors, and cables other than those specified may result in increased emission and/or decreased immunity and inaccurate readings of the tCOM+.

❗ **CAUTION:** To ensure accurate performance and prevent device failure, do not subject the tCOM+ to heavy moisture, such as direct exposure to rain. Such exposure may cause inaccurate performance or device failure. If the tCOM+ becomes wet accidentally, it should be removed from mains, wiped dry externally, allowed to dry thoroughly, and inspected by qualified service personnel before further use.

- **CAUTION:** Do not spray, pour, or spill any liquid on the tCOM+, its accessories, connectors, switches, or openings in the chassis. If the tCOM+ has been wetted accidentally, it must be removed from AC power, wiped dry externally, allowed to dry thoroughly, and inspected by qualified service personnel before further use.
- **CAUTION:** Do not lift the tCOM+ by any cable because they could disconnect from the tCOM+, causing the tCOM+ to fall on the patient.
- **CAUTION:** Do not sterilize any parts of the SDMS by irradiation, steam, ethylene oxide, or H_2O_2 plasma.
- **CAUTION:** Use only the external power supplies provided by Sentec and described in the tCOM+ Instruction Manual.
- **CAUTION:** Do not connect the tCOM+ to an electrical outlet controlled by a wall switch, because the tCOM+ may be accidentally turned off once the battery is depleted.
- **CAUTION:** If the monitor is operated on an external power supply with a depleted battery and the connection to mains is subsequently lost, the monitor will shut down immediately.
- **CAUTION:** When connecting/ mounting the tCOM+ to accessory equipment (e.g., PCs, PSG-Systems, (wireless) networks, roll stands, mounting plates, incubators, etc.)), verify proper operation before clinical use of the tCOM+ and accessory equipment. In certain cases, it may be required that the accessory equipment must be connected to a grounded AC outlet. In case of doubt, consult qualified technicians.
- **CAUTION:** Ensure that the fan of the tCOM+ is clear of any obstructions and that the tCOM+ is in a well-ventilated dust-free atmosphere. Failure to do so could cause damage or malfunction of the tCOM+.

Note: The tCOM+ monitor is not intended for diagnosis; it is intended only as an adjunct in patient assessment. It must be used in conjunction with clinical signs and symptoms. The tCOM+ is a cutaneous blood gas monitor and not a blood gas analyzer.

Note: The SDMS only can be used in patients undergoing hyperbaric therapy if the tCOM+ remains outside the hyperbaric environment.

Note: The tCOM+ is protected against electrostatic/ defibrillator discharge. Parameter display may be temporarily affected during discharge/ defibrillation but will rapidly recover. Precisely follow the instructions given in the defibrillator manual.

Note: This device has been tested and found to comply with the requirements for medical devices according to the IEC 60601-1-2, and the Medical Device Regulation (EU) 2017/745. These requirements are designed to provide reasonable protection against harmful interference in a typical medical installation.

Note: During normal operation (except transport), it is recommended that the monitor is always connected to mains through the external tCOM+ power supply.

Note: If used with the 3 pin (hospital) power supply, this equipment must be connected to a supply mains with protective earth. Ensure that power and protective ground lines are connected correctly. If case of doubt (e.g., as this may be the case during home use of the tCOM+) use the two pin (home) power supply.

Note: For US, respectively Japan: Grounding reliability can only be achieved when the tCOM+ is connected to an equivalent receptacle marked HG (Hospital Grade), respectively HGJ (Hospital Grade Japan).

Note: Certain types of mobile telecommunication equipment could potentially interfere with tCOM+ operation. Otherwise, degradation of the device performance could result.

Note: Accessory equipment (e.g., a PC) connected to the tCOM+'s data ports must be certified according to the IEC 60950 standard. All resulting combinations of equipment must be in compliance with the IEC standard 60601-1 systems requirements. Anyone who connects accessory equipment to the tCOM+ configures a medical system and is, therefore, responsible for ensuring that the resulting system complies with the requirements of standard IEC 60601-1 and the electromagnetic compatibility standard IEC 60601-1-2.

Note: To maintain monitor readiness in-between monitoring if using a Sentec TC Sensor:

- **ALWAYS** clean the sensor before putting it into the Docking Station.
- **ALWAYS** store the clean sensor in the Docking Station.
- **ALWAYS** keep the monitor switched on.

Note: Do not immerse the tCOM+ nor the connectors of any connecting cables in liquid solution. Plugs and connectors meticulously must be always kept clean and dry.

Note: Ensure that the tCOM+ is properly grounded when operating connected to mains through the external power supply (applies only to hospital power supply). If you are uncertain whether the AC outlet is properly grounded, disconnect the tCOM+ from the external power supply and use the battery power. Contact a qualified electrician to examine the outlet for ground connections.

Note: When installing/setting up the tCOM+, ensure that the tCOM+ can easily be disconnected from the external power supply at any given time.

1.2.2 Safety Information regarding Service Operations

CAUTION: Chemicals from a broken LCD display panel are toxic when ingested. Use caution when handling a tCOM+ with a broken display panel. Electronic components may content toxic chemicals. Do not ingest chemicals from a broken electronic component.

CAUTION: Dispose of battery in accordance with local requirements and regulations.

Note: Maintenance and repairs requiring removal of the case or covers may only be carried out by Sentec authorized service personnel trained by Sentec AG or accredited partners.

Note: Use only accessories and spare parts supplied or recommended by Sentec. Do not perform other service and repair activities than specified and described by Sentec. Failure to comply may result in physical injury, inaccurate measurements, and/or damage to the device.

2 The Sentec Digital Monitoring System (SDMS)

2.1 The tCOM+ Patient Monitor

2.1.1 Front Panel

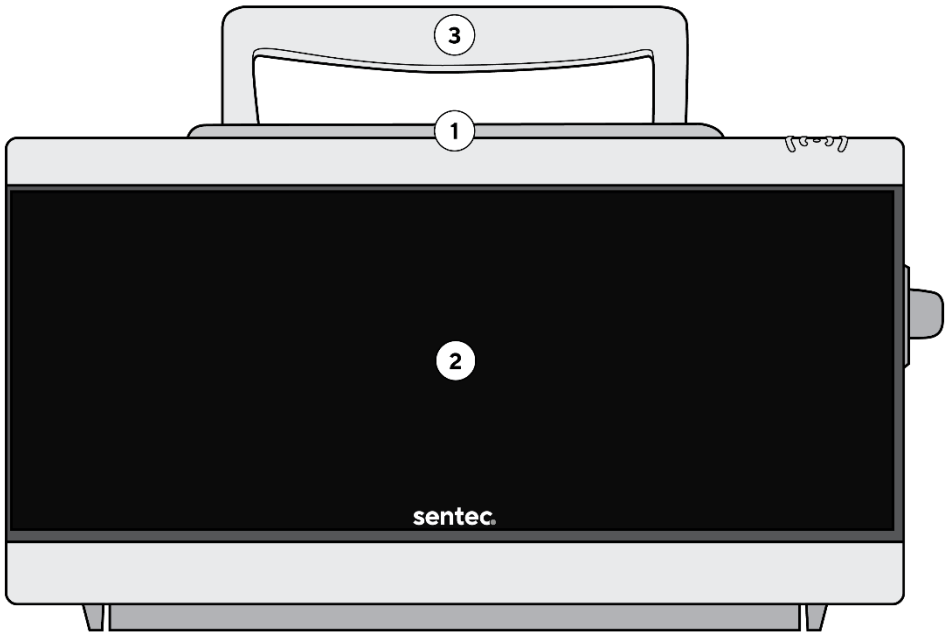


Figure 1 - Front Panel

1 LED Bar
2 Touchscreen
3 Handle

2.1.2 Side Panels



Figure 2 - Left Panel

4 Gas Bottle
5 DATA/SERVICE USB Port (USB C)
6 ON/OFF Button



Figure 3 - Right Panel

7 Docking Station

2.1.3 Rear Panel

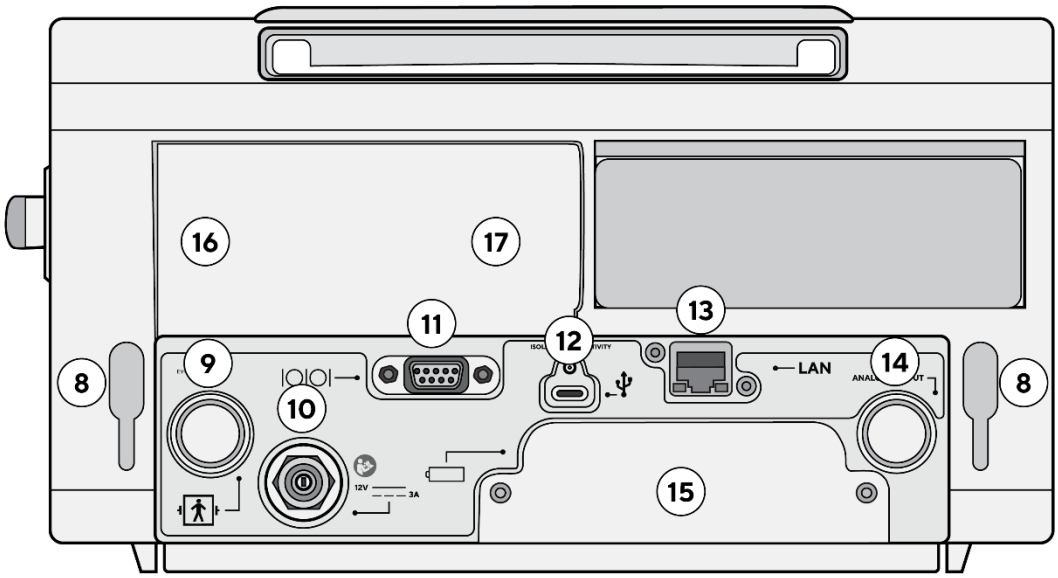


Figure 4 - Rear Panel

8 Cable Holder Slots
9 Sensor Connection Port
10 DC Power Connector
11 Serial Data Port (RS-232)
12 Isolated Connectivity Port (USB C)
13 Network Port (LAN)
14 Analog Output Port
15 Battery Cover
16 Fan
17 Speaker

2.1.4 Top View



3 Maintenance/Safety & Functionality Tests

At normal use there is no internal adjustment or new calibration of the tCOM+ required. However, to guarantee continuous performance, reliability, and safety of the SDMS routine checks and maintenance procedures (including cleaning/disinfection) as well as safety checks should be performed regularly.

Routine checks and maintenance procedures that can be performed by the **operator** are described in the INSTRUCTION MANUAL for tCOM+ Transcutaneous Monitor & Sensors (Chapter 4.1 “Routine checks, service, and maintenance procedures”).

Instructions for cleaning and/or disinfecting are provided in the INSTRUCTION MANUAL for tCOM+ Transcutaneous Monitor & Sensors (HBQ-179) and in ‘Cleaning and Disinfection Agents’ (HBQ-122).

It is recommended that **qualified personnel** perform the maintenance, and a complete safety and functionality test as specified in this manual at regular intervals (recommended every 12 months but at least every 24 months).

Note: Prior to maintenance or service all components of the SDMS must be cleaned and disinfected. If equipment is sent to service personnel, a local Sentec representative or to Sentec a completed *“Certificate for Disinfection”* (*see section* 8.1) must be attached.

Note: If any of the tests specified below fails, the respective problem must be fixed before the equipment is put back to operation.

3.1 Sentec TC Sensors

Verify that the sensor has been cleaned / disinfected as described in the tCOM+ Instruction Manual and “Cleaning and Disinfection Agents” (HB-010143). If necessary, remove dried up gel / electrolyte residues with a soft brush while immersing the (membraned) sensor into clean tap water.

3.1.1 Cleaning and Disinfection

Instructions for cleaning and/or disinfecting are provided in the INSTRUCTION MANUAL for tCOM+ Transcutaneous Monitor & Sensors (HBQ-179) and in ‘Cleaning and Disinfection Agents’ (HBQ-122).

3.1.2 Visual Inspection

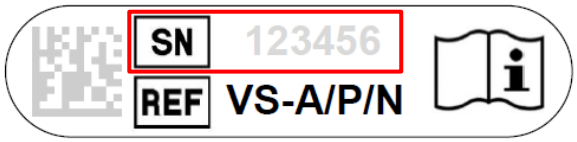
Visually inspect the sensor body, cable, and membrane for damage. Check the cable plug and its connectors for damages.

If any of the components of the sensor is irreversibly damaged, replace the sensor with a new Sentec TC Sensor.

3.1.3 Labeling

Check presence/legibility of the label on the plug of the sensor cable.

The sensor’s serial number can be found on the label on the plug (see figures below), either after the SN symbol (left figure) or after the application identifier (21) (right figure).



SN label with serial number (123456) after the SN symbol



UDI label with serial number (123456) after the application identifier (21)

If the label is not present / legible then connect the sensor to a tCOM+, switch the monitor on and read the sensor's serial number in the menu "System information".

3.1.4 Sensor LEDs

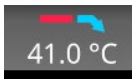
Connect the sensor to the tCOM+ and switch on the monitor. Verify that the sensor emits red light and is recognized by the tCOM+. When using an OxiVenT™ Sensor additionally verify that the sensor also emits green light (provided PO_2 is activated).

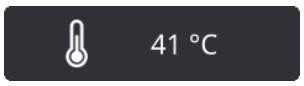
Note: To be able to verify the function of the green LED, a profile with PO_2 activated must be selected.

Then insert the sensor into the Docking Station. Verify that the calibration screen activates.

If the above-described functions do not operate properly, refer to section 4.3.1 *"Troubleshooting PCO2 / PO2 monitoring"* to evaluate the problem.

3.1.5 Sensor Temperature Display

Tap on the temperature indicator 

in the status bar to open the temperature settings and check if the set temperature 

matches the actual sensor temperature (shown by the indicator) within ±0.3 °C.

3.1.6 Sensor Temperature Surveillance

Note: The 'Sensor Temperature' Icon in the Status Bar indicates the measured sensor temperature (in °C).

Connect the sensor to the tCOM+ and switch on the monitor. Verify that the sensor temperature reaches the SET temperature within 2 to 3 minutes. Then immerse the sensor into clean tap water of 45 °C – 50 °C and verify that the measured sensor temperature increases (i.e., by observing the temperature indicator located in the Status Bar of the tCOM+ display). Verify that the red sensor LED switches off, a low priority alarm starts to sound and that the message "Temp. limiter active" (or 'Sensor problem 38', 'Sensor problem 42') displays.

If the message "Sensor Fault xx" or "SFxx" was triggered, you must restart the tCOM+ to reset the message. Do not use the sensor if the message "Sensor Fault xx" or "SFxx" is not reset by power cycling the tCOM+. Instead contact qualified service personnel or your local Sentec representative.

Do not use the sensor if the test fails!

CAUTION: Ensure that the temperature of the water bath is < 50 °C. Temperatures above 50 °C may damage the sensor.

3.1.7 Cleaning / Inspecting without Membrane

1. Remove the sensor membrane using the membrane remover located at the bottom of the Membrane Changer as shown in the figure below (note that the Membrane Changer is shown upside down). Step 1: Slide the Sentec TC Sensor into the membrane remover with its membrane facing the bottom of the Membrane Changer. Step 2: Lift the Sentec TC Sensor up to remove the membrane from the sensor body.

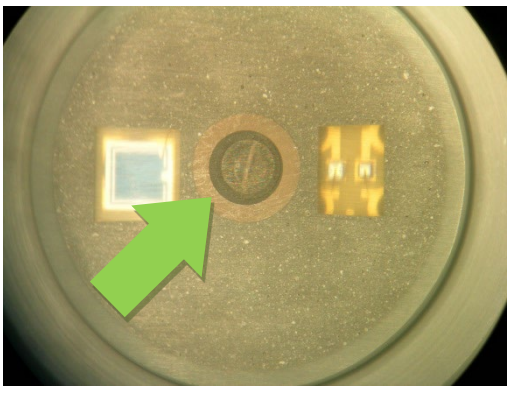


Figure 5 Removal of the Membrane from the TC Sensor

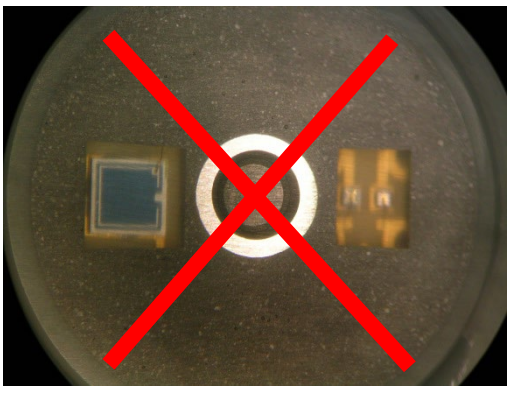
- **CAUTION:** Do not touch any of the measuring units in the center of the sensor surface after removing the sensor membrane. Do not rub the sensor surface.
- **CAUTION:** Do not leave the sensor without membrane in air. Perform the following steps uninterrupted.

2. Immerse the sensor into clean, room temperature tap water for 3 minutes.
3. Use a soft brush to completely remove dried up gel / electrolyte residues from the grooves on the sensor circumference.
4. Softly rinse the sensor with clean tap water.
5. Dab the sensor dry using a clean lint-free towel. Pay attention not to touch the ring and the pH-glass in the center of the sensor surface. Do not rub the sensor surface!
6. Inspect the ring around the pH-glass for damages and brownish color (*see Figure 6 Top view of the V-Sign™ Sensor*). Note that the same criteria apply for the OxiVenT™ Sensor.

If the ring around the pH-glass is broken, parts of it are missing, the brown color is lost, or the ring has a metallic luster, then replace the sensor with a new Sentec TC Sensor.



The sensor has an intact brown ring around the pH-glass.



The ring around the pH-glass is intact but has a metallic luster, this sensor needs to be replaced

Figure 6 Top view of the V-Sign™ Sensor

7. Inspect the sensor to ensure that the grooves on the sensor circumference are clean and intact. **If the grooves on the sensor circumference are damaged replace the sensor with a new Sentec TC Sensor. Do not use** the sensor if there is any visible damage to/trapped air underneath the membrane, or if the sensor's red LED does not light when the sensor is connected to the tCOM+.

8. OxiVenT™ Sensor only: Verify that the white oxygen sensing spot (i.e., off-centered, white, circular spot) is present and intact.



The white oxygen sensing spot is defect. This sensor needs to be replaced

Figure 7 Top view of the OxiVenT™ Sensor

9. If the visual inspections (steps 6 to 8) passed re-membrane the sensor using the Membrane Changer (for instructions, please refer to the Directions for Use of the Membrane Changer).

10. After placing a new membrane, verify that

    a) the membrane ring is intact and securely seated on the sensor.

    b) there are no air bubbles between membrane and sensor surface.

    In case of loose fit, damage, or trapped air, you must repeat the membrane change.

    If no faults are noted, confirm the membrane change in the menu of the monitor and calibrate sensor.

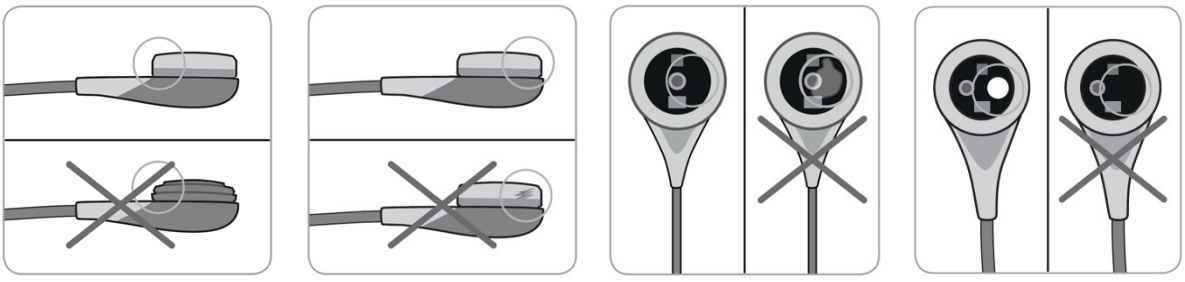


Figure 8 Inspection after re-membraning the sensor

3.1.8 PCO_2 / PO_2 Sensitivity

Note: If the procedure below is not strictly followed, the 'Sensitivity Test' will abort. Strictly observe the indicated times. 'Sensor Problem 12: Calibration Failed' (PCO_2 sensitivity) or 'Sensor Problem 72: PO_2 Calibration Failed' (PO_2 sensitivity) will only reset if the test was successful.

Note: If a V-Sign™ Sensor is connected to the tCOM+ the function 'Sensitivity Test' will test the sensor's PCO_2 sensitivity. If an OxiVenT™ Sensor is connected to the tCOM+ the function 'Sensitivity Test' will test the sensor`s PCO_2 and PO_2 sensitivity in parallel.

Test the PCO_2/PO_2 sensitivity as follows:

1. Make sure the Monitor used for the test has no gas leak in Docking station.
2. Insert the sensor into the Docking Station.
3. Open the Sensor Maintenance menu (e.g. by tapping on the Sensor icon in the status bar) and select Sensitivity Test. The Monitor will guide you through the following steps.
4. Open the Docking Station door within 1 minute to expose the sensor to ambient air.
5. After 2 minutes, the Status Message 'Close Docking Station door' and the corresponding image appears. Insert the sensor into the Docking Station and close it within 10 minutes.
6. The Monitor does a leak test and automatically proceeds to the next step.
7. Sensor is being calibrated. The Progress bar indicates the estimated remaining time.

If the 'Sensitivity Test' passed, the Status Message 'Ready for use' appears.

If the 'Sensitivity Test' for PCO_2 failed, 'Sensor Problem 12: Calibration Failed' will display when the sensor is in the Docking Station and PCO_2 subsequently will be flagged as invalid.

If the 'Sensitivity Test' for PO_2 failed, 'Sensor Problem 72: PO_2 Calibration Failed' will display when the sensor is in the Docking Station and PO_2 subsequently will be flagged as invalid.

If the calibration takes longer than 14 minutes 'PCO_2 slow' or 'Sensor Problem 11: Calibration Failed' or 'Sensor problem 'Sensor Problem 74: PO_2 Calibration Failed' is displayed.

Before repeating a failed test, please refer to troubleshooting P0304 and P0305 (see section 4.3.3 *"Sensor-specific troubleshooting"*).

3.1.9 Function Test: PCO_2 / PO_2

Calibrate the TC Sensor and apply it to the ear of a healthy person: Verify, that the PCO_2 readings equilibrate (stabilize) within 2 to 10 minutes and are within 33 to 47 mmHg after stabilization.

In case of an OxiVenT™ Sensor additionally test PO_2 function:

Power-cycle the tCOM+ (i.e., switch it off and on again). Then, calibrate the OxiVenT™ Sensor. After successful calibration the PO_2 function test can be performed as follows: Open the Docking Station Door to expose the sensor to ambient air. Then enable the 'Enforced Sensor On Patient Mode' to start measuring PO_2 (Menu Measurement settings / Start monitoring). Wait until PO_2 readings equilibrate (stabilize), usually within 2 to 10 minutes. Check the atmospheric pressure [Patm] (System Information / Barometer Value) and verify that PO_2 readings are within the range shown in the tables below. Note that PO_2 ranges in the tables below are only valid for ambient temperatures between 18°C and 28°C and relative humidity between 30% r.H. and 60% r.H.!

If these requirements are not met, refer to section 4.3.1 *"Troubleshooting PCO2 / PO2 monitoring"* to evaluate the problem.

Note: When using an OxiVenT™ Sensor, it is recommended to check the remaining 'Usage Time' / 'Lifetime'. This can be done on the Monitor's Menu "System Information" (refer to the tCOM+ Instruction Manual for further information).

PO_2 Function Test Control Table PO_2 – in mmHg and kPa

PO_2 ranges are valid for ambient temperatures between 18 °C and 28 °C and relative humidity between 30% r.H. and 60% r.H.

Patm [mmHg]	ambient_ min	ambient PO_2	ambient_ max	Patm [kPa]	ambient_ min	ambient PO_2	ambient_ max
600	116	123	130	80.0	15.4	16.4	17.4
605	117	124	131	80.7	15.6	16.6	17.5
610	118	125	133	81.3	15.7	16.7	17.7
615	119	126	134	82.0	15.9	16.9	17.8
620	120	128	135	82.7	16.0	17.0	17.9
625	121	129	136	83.3	16.1	17.1	18.1
630	122	130	137	84.0	16.3	17.3	18.2
635	123	131	138	84.7	16.4	17.4	18.4
640	124	132	139	85.3	16.6	17.6	18.5
645	125	133	140	86.0	16.7	17.7	18.6
650	126	134	141	86.7	16.8	17.8	18.8
655	127	135	142	87.3	17.0	18.0	18.9
660	128	136	143	88.0	17.1	18.1	19.1
665	129	137	144	88.7	17.3	18.3	19.2
670	130	138	145	89.3	17.4	18.4	19.3
675	132	139	146	90.0	17.5	18.5	19.5
680	133	140	147	90.7	17.7	18.7	19.6
685	134	141	148	91.3	17.8	18.8	19.8
690	135	142	149	92.0	18.0	19.0	19.9
695	136	143	150	92.7	18.1	19.1	20.0
700	137	144	151	93.3	18.2	19.2	20.2
705	138	145	152	94.0	18.4	19.4	20.3
710	139	146	153	94.7	18.5	19.5	20.5
715	140	147	154	95.3	18.6	19.6	20.6
720	141	148	156	96.0	18.8	19.8	20.7
725	142	149	157	96.7	18.9	19.9	20.9
730	143	151	158	97.3	19.1	20.1	21.0
735	144	152	159	98.0	19.2	20.2	21.2
740	145	153	160	98.7	19.3	20.3	21.3
745	146	154	161	99.3	19.5	20.5	21.4
750	147	155	162	100.0	19.6	20.6	21.6
755	148	156	163	100.7	19.8	20.8	21.7
760	149	157	164	101.3	19.9	20.9	21.8
765	150	158	165	102.0	20.0	21.0	22.0
770	151	159	166	102.7	20.2	21.2	22.1

3.1.10 Function Test: SpO_2 + PR

Choose an adult Profile with SpO_2 channel enabled. Apply the sensor to the earlobe of a healthy person: Compare SpO_2 and PR readings against the readings of a reference pulse oximeter (e.g., N595 with Durasensor 100 from Nellcor, or another SDMS with V-Sign Sensor 2 or OxiVenT Sensor). The SpO_2 and PR reading should be within ± 3% SpO_2 and ± 3 bpm, respectively.

If this requirement is not met, refer to section 4.3.1 *"Troubleshooting SpO2 / PR monitoring"* to evaluate the problem.

3.2 The tCOM+ Patient Monitor

3.2.1 Cleaning and Disinfection

Clean or disinfect the tCOM+ as often as required per institutional ordinances (recommended weekly). Instructions for cleaning and/or disinfecting are provided in the INSTRUCTION MANUAL for tCOM+ Transcutaneous Monitor & Sensors (HBQ-179) and in 'Cleaning and Disinfection Agents' (HBQ-122).

3.2.2 Visual Inspection

Visually check the entire tCOM+ for possible damages.

If damage is detected, the tCOM+ needs to be taken out of operation immediately and either have service parts replaced according to section 5 *"Replacement of Spare Parts"* or sent to Sentec for repair/replacement.

3.2.3 Labeling

Check the tCOM+ for presence and legibility of the following labels and symbols.
If the below requirements are not met contact your local Sentec representative or send back the monitor according to section 8.1 *"Instructions for Shipment"*.

3.2.3.1 Product Label

Label with variable and static parts (see Figure 9)

- Variable part: SN, REF, , UDI
- Static part: CE, manufacturing address, name, logo, certification information

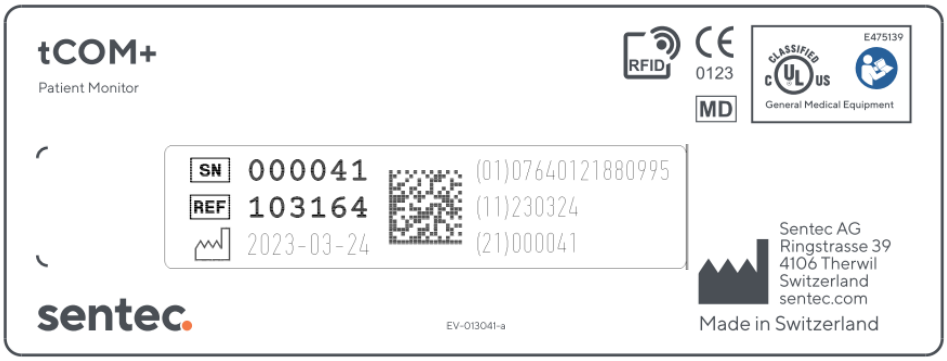


Figure 9 Product label

3.2.3.2 Interface Label

Label of ports at the back (see Figure 10)

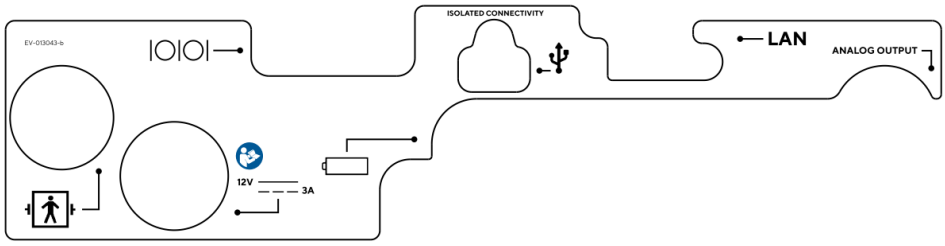


Figure 10 Interface label

3.2.3.3 Bottom Label

Label of additional information of product certification and regulation (see Figure 11)

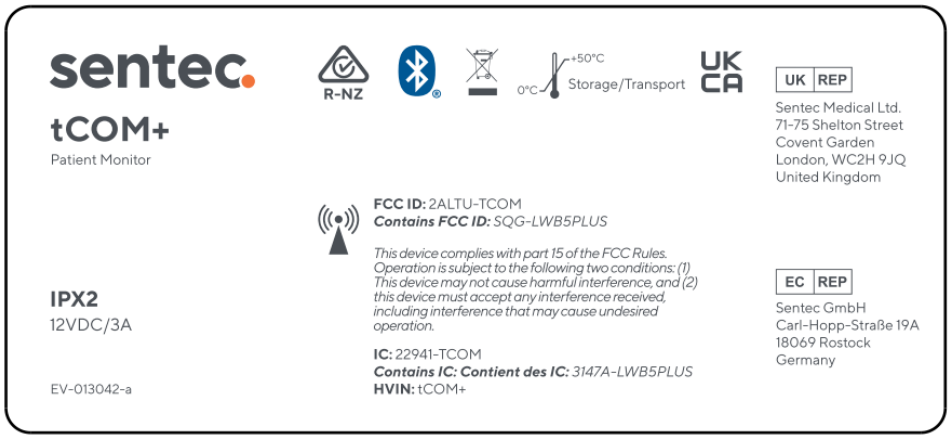


Figure 11 Bottom label

3.2.3.4 Country Label

Some countries require specific labeling which are subject to change. Due to their large variety, they are not shown here in detail.

3.2.4 Electrical Safety Check

The Sentec Digital Monitoring System (SDMS) has been designed to comply with the standards IEC 60601-1, ANSI/AAMI ES60601-1 and CAN/CSA-C22.2 No.60601-1 (UL and cUL). The sensor is an applied part of Type BF. The interface ports provide 2 MOOP, except the Ethernet port (1 MOOP). A detailed overview is provided in the Instruction Manual, Chapter ‘Means of Isolation (mains)’.

To ensure electrical safety of the unopened equipment it is recommended that **qualified service personnel** (must be familiar with the standards applicable to the institution and the country) performs a electrical safety check every 12 months (recommended) or 24 months (mandatory) according to IEC 62353 for instruments classified as Class II.

Requirements: The tCOM+ was not opened before and the safety seal is intact. Otherwise contact your local Sentec representative.

Tools: Electrical safety analyzer for tests according to IEC 60601-1 / IEC 62353 (e.g., Gossen Metrawatt Seculife ST Pro). If the required equipment is not available, contact your local Sentec representative.

Note: Before performing the safety test, read the operating instructions and make sure you are familiar with the usage of your safety analyzer.

Note: Make sure that all connecting cables are plugged in correctly. Poor or missing connections may lead to false test results.

The following connections must be tested. If the specifications are not met, the tCOM+ needs to be taken out of service. Contact your local Sentec representative.

Note: Due to the Class II design with the external power supply unit protective earth resistance is not applicable.

3.2.4.1 Visual Check

Check the AC power adapter for damage of the isolation. Check the connectors for damage.

3.2.4.2 Insulation Resistance (AC mains adaptor)

Must be more than 7 MΩ @500V DC:

- AC mains adaptor L+N (shorted) to DC Plug [–] (outer sleeve or thread if plugged)

3.2.4.3 Patient Insulation Resistance

Must be more than 70 MΩ @500V DC. Make sure to ground the DC receptacle [–] (thread) if the used tester does not support point-to-point measurement.

- DC receptacle [–] (thread) to Sensor Port (all pins shortened)
- DC receptacle [–] (thread) to Isolated Connectivity USB Port (shield)

Note: This test is not mandatory. Test execution shall be considered if there is any doubt regarding the insulation of the equipment. See also IEC 62353:2014.

3.2.4.4 Equipment Leakage Current

Applicable only for the hospital configuration (tCOM+ in combination with the hospital grade AC mains adaptor with functional earth). The test can be performed according to the method described in IEC 62353:2014. Alternatively, the test can be performed according to IEC 60601-1 AMD 2:2020 Figure 13 considering FE connection as PE.
Perform the following test cases:

- Equipment leakage current of the Sensor Port (all pins shortened)
- Equipment leakage current of the Isolated Connectivity USB Port (shield)

The current must be below 500uA, except for differential method the limit is 100uA.

3.2.4.5 Patient Leakage Current

The test must be performed according to the method described in IEC 62353:2014, Fig. 10, Class II, with the monitor turned on.

Perform the following test cases in either position of the mains plug (polarity):

- Applied part leakage current of the Sensor Port (all pins shortened)
- Applied part leakage current of the Isolated Connectivity USB Port (shield)

The applied part leakage current must be lower than 500uA.

3.2.5 POST

After turning on the tCOM+ performs a power-On Self-Test (POST), which tests the monitor's circuitry and functions (internal test). The display, as well as the visual and audible indicators (i.e., speaker, buzzer, LED bar) are activated.

The result of the POST is indicated by "POST passed" (may not be displayed or very short) or "POST failed" message.

If no problems are detected during the POST, the monitor may shortly show "POST passed" before proceeding to the main screen; otherwise, the monitor will stop at the failed test and display an error message. Try to restart the monitor and if the problem persists, refer to P0405 (see section 4.3.4 *"tCOM+ specific troubleshooting"*).

3.2.6 Power Button

1. Press the power button on the left of the monitor.
2. The menu "Shut Down" will appear on the screen. Close this screen by selection X on the top right corner.

If the above-described process does not operate properly, contact your local Sentec representative.

3.2.7 Clock Settings

Verify that the clock settings are correct. If necessary, adjust the clock in the tCOM+ Menu "Advanced Settings → Date / Time" (a password may be required).

Power cycle the tCOM+ and verify the clock settings.

If the clock settings are not correct, refer to P407 in section 4.3.4 *"tCOM+ specific troubleshooting"* to evaluate the problem.

3.2.8 Display

Check the display for defects. Check the readability of the following critical areas: graph window, numerical displays, status bar.
If the above-described functions do not operate properly contact your local Sentec representative.

3.2.9 Buzzer / Speaker (POST)

Switch on the tCOM+ and during POST verify that you hear a short, high-pitched beep from the buzzer followed by three low-pitched tones from the speaker.

Buzzer and Speaker are on different Modules. If one or both are not audible, refer to section 4.3.4 *"tCOM+ specific troubleshooting"* to evaluate the problem.

3.2.10 Analog Output

Usage of the analog output (setting up the channels and ranges for PSG) is described in the tCOM+ instruction manual.

This test is only required if the analog output is used, and the port is suspected of not working properly. It will use the monitor's internal calibration routine to verify min. and max. voltage levels, independently from custom channels and range assignment.

Required Tools:

- Digital multimeter
- tCOM+ PSG Cable X (*available from Sentec*).

Use the multimeter in V DC mode to measure all channels against signal ground on the open-end wires of the PSG Cable X, which are labeled with the default Signals, to identify the channels (if no PSG cable X is available, refer to the annex for the pinout).

Channel	Wire label (Default signal)
CH 0	Pleth
CH 1	PR
CH 2	SpO_2
CH 3	PCO_2
All	Signal Ground

In the menu "Advanced Settings > Interfaces > Analog / PSG" (a password may be required) tap on Calibrate Channels.

Output: 1.0 V → Verify with multimeter for all channels (0.95 ... 1.05 V_{DC})

Tap "Confirm Voltage"

Output: 0.0 V → Verify with multimeter for all channels (-10 ... +10 mV_{DC})

Tap "Confirm Voltage"

If the above-described functions do not operate properly, refer to section 4.3.4 *"tCOM+ specific troubleshooting"* to evaluate the problem.

3.2.11 Serial Data Port (RS-232)

For a detailed description of the Serial Data Port (RS-232) refer to the corresponding section of the tCOM+ instruction manual.

This test is only required if the RS-232 Port is used, and the port is suspected of not working properly in its intended use (e.g. connection to a third-party device).

1. Connect your PC (using a USB-serial adapter) to the tCOM+
2. Start a simple terminal program e.g. HTerm (available at https://hterm.org/)
3. Make sure the Monitor's protocol is set to «SentecLink» and «115200»
4. Make sure to disable the serial encryption (note the setting before and restore after the test)
5. Set Hterm for the respective COM-Port and baud rate to 115200.
6. Set Hterm for «Send on enter CR-LF»
7. Send «Online>>» , so the monitor will send a status message every second.
8. Send «Online<<» to stop the status message output.
9. Make sure to enable the serial encryption again, if it was enabled before the test.

If the above-described functions do not operate properly, the tCOM+ needs repair. Refer to section 4.3.4 *"tCOM+ specific troubleshooting"* to evaluate the problem.

3.2.12 Capacity of Rechargeable Lithium Battery

This test sequence can take up to 4 hours, make sure to allocate enough time for charging and/or discharging.

1. Make sure the battery is fully charged and the display shows 100%.
2. Connect a properly working sensor and place it into the calibration chamber.
3. Set the display brightness to maximum (Display Menu > Brightness).

4. Disconnect the tCOM+ Monitor from mains power supply, start a timer and check if the monitor is still running after the typical needs in your organisation. Sentec recommends at least the following time period:

If constantly connected to Mains:	>30 minutes
If used for Transport:	Typical transport duration × 1.5
General recommendation, if unknown:	3 hours

Alternatively, if you need to know the full capacity of the battery, you need to let it run out of power. To make sure the shutdown is not missed, it's recommended to record the online output via serial port using a terminal program (acc. to chapter 3.2.11). Make sure the connected computer is mains powered and power saving options are disabled so the terminal program can record uninterrupted.

If the battery does not meet the required capacity, refer to section 5.8 *"Replacement, Battery"*.

3.2.13 Alarm by Display

Initiate a low priority alarm (e.g., by disconnecting the Sentec TC Sensor from the tCOM+). Verify that the corresponding status message is displayed in the status bar.

If the above-described functions do not operate properly, the tCOM+ needs repair. Refer to section 4.3.4 *"tCOM+ specific troubleshooting"* to evaluate the problem.

3.2.14 Alarm by Speaker

Initiate a low priority alarm (e.g., by disconnecting the Sentec TC Sensor from the tCOM+). Verify that the audible component of a low priority alarm is audible (one burst of two pulses (1 dyad) repeated every 15 seconds).

If the above-described functions do not operate properly, the tCOM+ needs repair. Refer to section 4.3.4 *"tCOM+ specific troubleshooting"* to evaluate the problem.

3.2.15 Alarm by LED Bar

Initiate a low priority alarm (e.g., by disconnecting the Sentec TC Sensor from the tCOM+). Verify that the LED bar lights up cyan evenly.

If the above-described functions do not operate properly contact your local Sentec representative or send back the monitor according to *section* 8.1 *"Instructions for Shipment"*.

3.3 Docking Station

3.3.1 Cleaning and Disinfection

Clean or disinfect the Docking Station (including the Docking Station gasket) as often as required per institutional ordinances (recommended weekly) using a cotton swab moistened with 70% isopropanol.

3.3.2 Visual Inspection

Inspect the Docking Station door and the gasket for mechanical damages. The closing mechanism must lock properly. Also check the Gasket for the Gas bottle for deformation or cracks.

If you detect a possible damage, replace the defective parts according to section 5 *"Repairs"*.

Note: Replacement of both gaskets is recommended annually.

3.3.3 Gas Level Indicator

Tools
- 1 full Calibration Gas bottle

Check the functionality of the gas level indicator using a new gas bottle.

1. Fit a new gas bottle and switch the tCOM+ ON.
2. Insert the sensor in the Docking Station and verify that the "100% gas icon" displays in the status bar.
3. Remove the gas bottle from the tCOM+ and verify that the yellow "0% gas icon" and and the message “Gas bottle empty” is shown in the status bar.

If the above-described functions do not operate properly, the tCOM+ needs repair. Refer to section 4.3.4 *“tCOM+ specific troubleshooting"* to evaluate the problem.

3.4 Sensor Adapter Cable

3.4.1 Cleaning and Disinfection

Clean or disinfect the `Sensor Adapter Cable´ as often as required per institutional ordinances (recommended weekly). For cleaning and disinfection instructions refer to the tCOM+ instruction manual).

3.4.2 Visual Inspection

Check the Sensor Adapter Cable for damages of the isolation. Check the Sensor Adapter Cable connectors for damages.

If you detect damages, the Sensor Adapter Cable needs to be replaced by a new Sensor Adapter Cable.

3.5 Opening the Membrane Changer

If the sensor is blocked within the Membrane Changer at any position, it can be opened as follows:

Identify the 6 latches holding together top and bottom part of the Membrane Changer together (see Figure 13 Opening the Membrane Changer). Insert a screwdriver behind the latches and use it as leverage to break out the 6 latches.

CAUTION: Wear protective goggles while breaking out the 6 latches.

CAUTION: This procedure will irreversibly destroy the Membrane Changer, but not harm the sensor if executed with care.

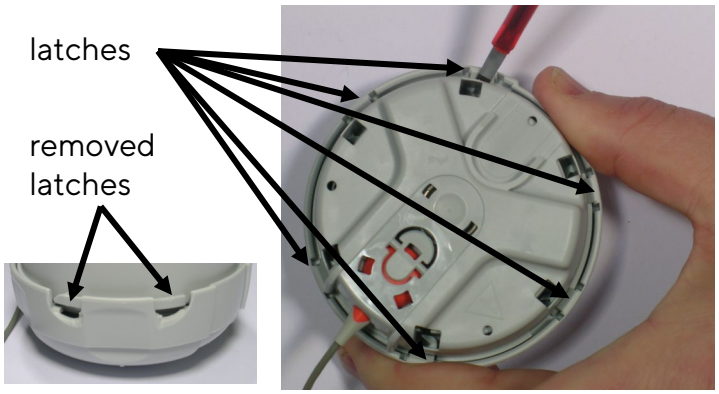


Figure 12 Opening the Membrane Changer

4 Troubleshooting

This section describes problems, possible causes and recommended corrective actions. Replacement of Service Parts (without opening the tCOM+ cover) are described in section 5.

Repairs that require opening the cover of the tCOM+ are to be performed by Sentec only.

Prior to maintenance or service or all components of the SDMS must be cleaned and disinfected. If equipment is sent to service personnel, a local Sentec representative or to Sentec a completed *"Certificate for Disinfection"* (*see section* 8.1) must be attached.

If you need to return devices for service or repair, follow the shipment instructions in *section* 8.1 *"Instructions for Shipment"*.

Send defective parts together with the completed "Device Service Request" to your local Sentec representative. The "Device Service Request" may be copied from section 8.1 *"Forms"*.

4.1 How to use the troubleshooting list

The troubleshooting list describes possible defects of the device. The problems are numbered (PXXXX) unambiguously. Please reference the respective PXXXX code whenever applicable.

The troubleshooting list comprises two levels of recommended actions to be performed

[1] by the **operator** (not shaded) and, if the problem persists,

[2] by a **qualified technician** (shaded in light grey), i.e., troubleshooting at service level (according to this manual) and, if the problem persists, the local Sentec representative needs to be contacted.

Note: Consider that the operator might not have performed the corrective actions recommended at operator-level and/or might not have identified the problem correctly. Therefore, check other possible causes and problems.

4.2 Downloading Data Logs

In most cases of a persisting problem that requires contacting the Sentec representative, it will be required to download the data Logs from the Monitor in question. It is recommended to download this beforehand and include with the service request as well as a precise description of where and when the problem occurred and what was already done.

To download the Data Logs, connect a USB Stick to the DATA/SERVICE USB Port on the left side of the Monitor. Go to the Review & Export Menu / Export Data Logs and follow the instructions on screen. The process will take several minutes. Data Logs can also be downloaded on the POST screen, if it failed.

4.3 Troubleshooting List

This section describes problems, possible causes and recommended actions for troubleshooting.

Note: Shadings in the troubleshooting table reflect different levels of recommended actions, i.e. for the operator (unshaded) and qualified service personnel (shaded light grey), *see section 4.1 "How to use the troubleshooting list"* for details.

4.3.1 Troubleshooting PCO_2 / PO_2 monitoring

When comparing PCO_2/ PO_2 values displayed by the tCOM+ with $PaCO_2$/ PaO_2 values obtained from arterial blood gas (ABG) analysis, pay attention to the following points:

- Carefully draw and handle blood samples.
- Blood sampling should be performed in steady state conditions.
- The $PaCO_2$/ PaO_2 value obtained from ABG analysis should be compared to the tCOM+'s PCO_2/ PO_2 reading at the time of blood sampling.
- In patients with functional shunts, the sensor application site and the arterial sampling site should be on the same side of the shunt.
- If the 'Severinghaus Correction Mode' (advanced settings menu) is set to 'Auto', the PCO_2 values displayed by the tCOM+ are automatically corrected to 37 °C (regardless of the patient's core temperature). When performing the ABG analysis, be sure to properly enter the patient's core temperature into the blood gas analyzer. Use the blood gas analyzer's '37 °C-$PaCO_2$' value to compare with the tCOM+'s PCO_2 value.
- Verify proper operation of the blood gas analyzer. Periodically compare the blood gas analyzer's barometric pressure against a known calibrated reference barometer.

Note: This section only is applicable for Sentec TC Sensors with PCO_2 and/ or PO_2 enabled.

Table 2 Troubleshooting PCO_2/ PO_2 Monitoring

Problem	Possible Cause(s)	Recommended corrective action(s)
P0100 After sensor application or a 'TC artifact' PCO_2 values increase slowly or PO_2 values decrease slowly or do not stabilize within 10 minutes	Sensor loosely attached to patient (air between sensor and skin; no contact liquid)	Verify that the sensor is securely attached and ensure that air gaps are eliminated between the skin and the sensor. If needed, reapply the sensor (on same or an alternate site). For instructions refer to the tCOM+ instruction manual. Consider using the Staysite™ Adhesive (model SA-MAR) to complementary attach Multi-Site Attachment Rings if more secure sensor attachment is required, e.g., in high humidity environments, for patients who perspire profusely and/or in challenging patient motion conditions. **Note:** Typically, $PaCO_2$ is > 25 mmHg in humans and PCO_2 of ambient air < 0.5 mmHg. If the contact between the sensor and skin is not hermetically sealed PCO_2 readings, consequently, may increase slowly after sensor application or a 'TC artifact' and be too low thereafter. **Note:** Typically, PaO_2 is < 100 mmHg in humans and PO_2 of ambient air > 130 mmHg. If the contact between the sensor and skin is not hermetically sealed PO_2 readings, consequently, may decrease slowly after sensor application or a 'TC artifact' and be too high thereafter. **Note:** Also refer to Status Messages 'Check sensor application', 'PCO_2 stabilizing', 'PO_2 stabilizing', or 'PCO_2/ PO_2 stabilizing' (refer to the tCOM+ instruction manual).
	Sensor applied to a measurement site that is inadequate /	Apply sensor to a recommended measurement site with an appropriate Sensor Attachment Accessory.

Problem	Possible Cause(s)	Recommended corrective action(s)
	not recommended for PCO_2/ PO_2 monitoring	For instructions refer to the tCOM+ instruction manual.
	Excessive use of contact liquid	Remove sensor. Clean sensor and measurement site with an alcohol pad. Reapply the sensor using only one drop of contact liquid.
P0101 PCO_2 too low OR PO_2 too high	Sensor loosely attached to patient (air between sensor and skin; no contact liquid)	Refer to identical 'Possible Cause' in P0100 for recommended corrective action(s)
	Sensor applied to a measurement site that is inadequate/ not recommended for PCO_2/ PO_2 monitoring	Refer to identical 'Possible Cause' in P0100 for recommended corrective action(s)
	Arterio-venous shunts, e.g., ductus arteriosus.	Ensure that the sensor site and the arterial sampling site are on the same side of the shunt
	PCO_2 part of the sensor unstable (causing drift)	Allow the sensor to stabilize for an additional 1 – 2 hours, i.e., store the sensor in the Docking Station with the monitor being switched on (refer to the tCOM+ instruction manual for additional sensor stabilization related information). **Note:** After switching-on the monitor or after membrane change, it is recommended to store the sensor in the Docking Station with the monitor being switched on for the duration indicated by the information message 'Remaining Recommended Sensor Stabilization' on the "Ready for use" screen (see tCOM+ instruction manual).
	Wrong / expired calibration gas was used during last calibration	Verify that Sentec Calibration Gas is used. If expired replace with a new gas bottle!
	Wrong / unstable barometric pressure during last calibration	Refer to P0408
	PCO_2 and or PO_2 sensitivity of the sensor low/ deteriorated	Refer to P0305
	PO_2 Measurement is influenced by high ambient light levels	Shield Sentec TC Sensor from strong ambient light. **Note:** Refer to Status Message 'High ambient light' (with Status Code 'SA').
		Clean/Inspect the sensor as described in section 3.1.7 *"Cleaning / Inspecting without Membrane"* and perform the 'Sensitivity Test' as described in section 3.1.8 *"PCO_2 / PO_2 Sensitivity & Calibration Duration"*.

Problem	Possible Cause(s)	Recommended corrective action(s)
		If problem persists, contact Sentec authorized service personnel.
	Significant PCO_2 drift	Verify that the sensor is not affected by loose fit of the membrane, trapped air under the membrane. If needed perform an unrequested membrane exchange (activate the menu "Membrane Change"). After membrane exchange calibrate the sensor and allow the PCO_2 part of the sensor to stabilize for 4 hours. If no re-membraning is needed allow the sensor to stabilize for an additional 2 – 3 hours.
	Sensor or tCOM+ defective	Contact your local Sentec representative.
P0102 PCO_2 too high OR PO_2 too low	Low perfusion at the (hypoperfused) measurement site	Limitation of the method**. Evaluate the patient status**. Shock, low cardiac index, hypothermia, and inadequate local vasodilatation are physiological limitations of the method. Avoid any mechanical pressure / stress on the measurement site. Verify that the **sensor temperature** is appropriately selected for the specific patient/ skin condition at the measurement site and the enabled parameters. When selecting/ increasing sensor temperature carefully balance benefit (more accurate measurements) versus risk (potential skin burns). Carefully read the section "Sensor Temperature & Site Timer" in the tCOM+ instruction manual. **Remember**: In most patients PO_2 monitoring requires higher temperatures than PCO_2 monitoring. In newborns, PO_2 correlates with PaO_2 almost in a one-to-one relationship at a sensor temperature of 43 to 44 °C, whereby the accuracy of PO_2 compared to PaO_2 is best up to PaO_2 = 80 mmHg (10.67 kPa), above which it increasingly tends to read lower than PaO_2 (especially in adults).
	Sensor applied to a measurement site that is inadequate / not recommended for PCO_2 monitoring.	Refer to identical 'Possible Cause' in P0100 for recommended corrective action(s)
	Arterio-venous shunts, e.g., ductus arteriosus.	Ensure that the sensor site and the arterial sampling site are on the same side of the shunt
	Presence of hyperoxemia (PaO_2 > 100 mmHg)	**Remember**: In newborns, PO_2 correlates with PaO_2 almost in a one-to-one relationship at a sensor temperature of 43 to 44 °C, whereby the accuracy of PO_2 compared to PaO_2 is best up to PaO_2 = 80 mmHg (10.67 kPa), above which it increasingly tends to read lower than PaO_2 (especially in adults).

Problem	Possible Cause(s)	Recommended corrective action(s)
	PCO_2 part of the sensor unstable (causing drift)	Refer to identical 'Possible Cause' in P0101 for recommended corrective action(s)
	Wrong / expired calibration gas was used during last calibration	Refer to identical 'Possible Cause' in P0101 for recommended corrective action(s)
	Wrong / unstable barometric pressure during last calibration	Refer to P0408
	Undetected gas leak in Docking Station.	Refer to P0502
	PCO_2 and or PO_2 sensitivity of the sensor low/ deteriorated	Refer to P0305
	PO_2 Measurement is influenced by high ambient light levels	Shield Sentec TC Sensor from strong ambient light. **Note:** Refer to Status Message 'High ambient light' (with Status Code 'SA') (see tCOM+ instruction manual).
		Clean/Inspect the sensor as described in section 3.1.7 *"Cleaning / Inspecting without Membrane"* and perform the 'Sensitivity Test' as described in section 3.1.8 *"PCO_2 / PO_2 Sensitivity"*. If problem persists, contact your local Sentec representative.
	Gasket dirty or defective	Clean or replace gasket according to section 5.4 *"Docking Station Gasket"*.
	Sensor or tCOM+ defective	Contact your local Sentec representative.
P0103 PCO_2 and/ or PO_2 suddenly change / change very fast ('TC artifact')	Sensor loosely attached to patient (air between sensor and skin; no contact liquid)	Refer to identical 'Possible Cause' in P0100 for recommended corrective action(s)
	Adhesive pad of Ear Clip / Multi-Site Attachment Ring does not stick any more	Reapply the sensor with a new Ear Clip / Multi-Site Attachment Ring. Pay attention not to wet the adhesive. For instructions, refer to the tCOM+ Instruction Manual. Consider using the Staysite™ Adhesive (model SA-MAR) to complementary attach Multi-Site Attachment Rings if more secure sensor attachment is required, e.g., in high humidity environments, for patients who perspire profusely and/or in challenging patient motion conditions.
	Electrocautery, HF surgery, defibrillation (mainly PCO_2 specific)	The values will come back to normal after electrocautery, HF surgery or defibrillation.

Problem	Possible Cause(s)	Recommended corrective action(s)
P0104 PCO_2 and/ or PO_2 changes very slowly	Sensor applied to a measurement site that is inadequate / not recommended for PCO_2 monitoring.	Refer to identical 'Possible Cause' in P0100 for recommended corrective action(s)
	Excessive use of contact liquid	Refer to identical 'Possible Cause' in P0100 for recommended corrective action(s)
	PCO_2 and/ or PO_2 part of the sensor is slow	Refer to P0304.

4.3.2 Troubleshooting SpO_2 / PR monitoring

Note: This section is applicable for Sentec TC Sensors (if SpO_2 / PR are enabled) and for Sentec's oximetry sensors.

Table 3 Troubleshooting SpO_2/ PR Monitoring

Problem	Possible Cause(s)	Recommended action(s)
P0200 SpO2 and / or PR too low or too high	Venous pooling (e.g., Trendelenburg position)	Limitation of the method. For placement of the sensor on the earlobe, forehead, or cheek: avoid conditions where the head is positioned lower than the heart.
	Some cardiovascular pathologies (e.g., causing cardiac arrhythmias)	Limitation of the method.
	Inadequate local blood flow at the measurement site	Evaluate the patient status. Shock, low cardiac index, inadequate local vasodilatation are physiological limitations of the method. Verify the sensor application and the appropriateness of the monitoring site. Avoid any mechanical pressure / stress on the sensor or measurement site. Secure the sensor cable with adhesive tape and a clothing clip. **Note:** Refer to the Status Message 'SpO_2 low signal' (see tCOM+ instruction manual).
	Electrocautery, HF surgery, defibrillation	The values will come back to normal after electro-cautery, HF surgery or defibrillation.
	Sensor applied to a measurement site that is inadequate / not recommended for SpO_2 monitoring.	Apply sensor to a measurement site recommended/ approved for SpO_2/ PR monitoring with an appropriate Sensor Attachment Accessory. For instructions refer to the tCOM+ instruction manual.
	Artifacts due to patient movements	Verify the sensor application and the appropriateness of the monitoring site. Secure the sensor cable with adhesive tape and a clothing clip. **Note:** Refer to the Status Message 'SpO_2 signal quality' (see tCOM+ instruction manual).

Problem	Possible Cause(s)	Recommended action(s)
	SpO_2/ PR Measurement is influenced by high ambient light levels	Shield Sentec TC Sensor from strong ambient light (e.g., lamps). **Note:** Refer to Status Message 'High ambient light' (with Status Code 'HA') (see tCOM+ instruction manual).
	Dysfunctional hemoglobin	Physiological limitation of the method. In patients with high carboxyhemoglobin levels (e.g., smokers) functional saturation measured with a pulse oximeter will overestimate functional saturation determined with a CO-oximeter.
	Excessive use of contact liquid causing light shunting	Remove the Sentec TC Sensor. Clean sensor and measurement site with an alcohol pad. Reapply the sensor using only one drop of contact liquid.
	Intravascular dyes	Limitation of the method. Dyes used as tracer, contrast medium, etc. may absorb light at the same wavelength as the one used for pulse oximetry measurements.

4.3.3 Sensor-specific troubleshooting

Table 4 Sensor-specific troubleshooting

Problem	Possible Cause(s)	Recommended action(s)
P0300 Despite a Sentec TC Sensor is connected to the tCOM+ the message 'Connect sensor' displays	Bad contact (cables not properly connected)	Verify connections (disconnect and reconnect all cables) and – if necessary – power cycle the tCOM+.
	Sentec TC Sensor and / or 'Digital Sensor Adapter Cable' defective	To identify the defective part, exchange the 'Sensor Adapter Cable' or the Sentec TC Sensor and try again (if necessary, permute and try on a different tCOM+). In case of damage of the Sentec TC Sensor replace it by original replacement parts. In case of damage of the 'Sensor Adapter Cable' replace it by original replacement parts.
	tCOM+ defective	If problem persists, contact qualified service personnel.
	tCOM+ defective	Refer to *section* 4.3.4 *"tCOM+ specific troubleshooting"* to evaluate the problem.
P0302 Sensor's red LED does not light (message 'Connect sensor' not displayed)	As a safety precaution sensor or tCOM+ did switch off all power consuming parts	Refer to error messages 'Temp. limiter active', 'SP38', 'SP42', 'SF39', 'SF43'. Remove auxiliary heat sources and allow to cool. If problem persists, contact your local Sentec representative.
	LED defective	Refer to error messages 'SP20: LED Failure' and 'SF21: LED Failure'. If problem persists, contact your local Sentec representative.

Problem	Possible Cause(s)	Recommended action(s)
P0303 Message 'SP10: Contact Service'	Sentec TC Sensor defective/ low voltage potential is detected	The Sentec TC Sensor is defective. Replace it with a new one.
P0304 PCO_2 slow (too slow) / Message 'SP11: Change sensor membrane/ SP14: Change sensor membrane' OR PO_2 slow (too slow) / Message 'Sensor problem 74'	Membrane related (defective or dried up electrolyte) PCO_2 electrode and/ or white PO_2 sensing spot contaminated	Change the membrane: - if it is damaged or missing, has a loose fit, or if there is trapped air or dry electrolyte under the membrane - sensor calibration takes almost 14 minutes or cannot be completed within 14 minutes **Note**: If the PCO_2 calibration takes 14 minutes the message 'PCO_2 slow' displays and PCO_2 subsequently will be flagged as questionable. **Note**: If the PCO_2 calibration cannot be finished within 14 minutes the message 'SP11: Change sensor membrane' displays and PCO_2 subsequently will be flagged as invalid. **Note**: If the calibration cannot be finished within approx. 30 minutes – due to failed extended calibration – the message 'SP14: Change sensor membrane' displays and PCO_2 subsequently will be flagged as invalid. **Note**: If the PO_2 calibration cannot be finished within 14 minutes the message 'Sensor problem 74' displays and PO_2 subsequently will be flagged as invalid. If after having changed the sensor membrane the calibration still takes almost 14 minutes or cannot be completed within 14 minutes, contact qualified service personnel to clean the sensor without membrane.
		Clean the sensor as described in *section 3.1.7 "Cleaning / Inspecting without Membrane"*. Then calibrate the sensor.
	Sentec TC Sensor defective	If the problem persists, the Sentec TC Sensor is defect. Replace it with a new one.

Problem	Possible Cause(s)	Recommended action(s)
P0305 PCO_2 sensitivity of the sensor low / Message 'SP12: Check membrane & gasket' OR PO_2 sensitivity of the PO_2 sensor low / Message 'Sensor problem 72'	Membrane related (defective or dried up electrolyte)	Change the membrane: - if it is damaged or missing, has a loose fit, or if there is trapped air or dry electrolyte under the membrane - if the 'Sensitivity Test' fails. Perform the 'Sensitivity Test' as described in *section 3.1.8 "PCO_2 / PO_2 Sensitivity & Calibration Duration*". **Note**: If the 'PCO_2 Sensitivity Test' fails the message 'SP12: Check membrane & gasket' displays and PCO_2 subsequently will be flagged as invalid **Note**: If the 'PO_2 Sensitivity Test' fails the message 'Sensor problem 72' displays and PO_2 subsequently will be flagged as invalid.
	PCO_2 electrode and/ or white PO_2 sensing spot contaminated	If after having changed the sensor membrane the 'Sensitivity Test' still fails, contact qualified service personnel to clean the sensor without membrane.
		Clean/Inspect the sensor as described in *section 3.1.7 "Cleaning / Inspecting without Membrane"* and perform the 'Sensitivity Test' as described in *section 3.1.8 "PCO_2 / PO_2 Sensitivity & Calibration Duration*".
	Sentec TC Sensor defective	Contact your local Sentec representative.
P0306 Status Message 'Temp. limiter active' / Message 'Sensor problem 38-42'	Temperature of the Sentec TC Sensor is increased, e.g., because of warming lamps or electric blankets	Shield sensor from heat.
	Sentec TC Sensor defective	If the problem persists, the Sentec TC Sensor is defect. Replace it with a new one.
	tCOM+ defective	Refer to section *4.3.4 "tCOM+ specific troubleshooting"* to evaluate the problem.
P0307 Sensor is blocked in Membrane Changer tool	Membrane change tool is blocked	Open the Membrane Change Tool as described in *section 3.5 "Opening the Membrane Changer"*.
P0308 Sensor calibration does not start	Sensor temperature is too low	Wait until the sensor temperature is less than 0.5 °C below the SET temperature.
		If problem persists, test the same sensor connected to another tCOM+ using another extension cable.
	Sentec TC Sensor defective	If the problem persists, the sensor is defect. Replace it with a new one.

Problem	Possible Cause(s)	Recommended action(s)
P0309 OxiVenT™ Sensor connected, but PO_2 measurement flagged to be invalid or not available	LED of the PO_2-module defective	The sensor can still be used for monitoring of the other parameters but PO_2. Replace OxiVenT™ Sensor if you wish to continue monitoring PO_2. **Note**: In this case, the message 'Sensor problem 70' will be displayed.
P0310 Message 'SP15: Change sensor membrane'	Membrane related (defective or dried up electrolyte)	Change the membrane: - if it is damaged or missing, has a loose fit, or if there is trapped air or dry electrolyte under the membrane - if the 'Sensitivity Test' fails Perform the 'Sensitivity Test' as described in section *3.1.8 "PCO_2 / PO_2 Sensitivity & Calibration Duration"*. **Note**: If the 'PCO_2 Sensitivity Test' fails, the message 'SP12: Check membrane & gasket' appears and PCO_2 is subsequently flagged as invalid. **Note:** If the 'PO_2 Sensitivity Test' fails, the message 'Sensor problem 72' appears and PO_2 is subsequently flagged as invalid.
	PCO_2 electrode and/or white PO_2 sensing spot contaminated	If, after having changed the sensor membrane, the 'Sensitivity Test' still fails, contact qualified service personnel to clean the sensor without membrane.
		Clean/inspect the sensor as described in *section 3.1.7 "Cleaning / Inspecting without Membrane"* and perform the 'Sensitivity Test' as described in *section 3.1.8 "PCO_2 / PO_2 Sensitivity & Calibration Duration"*.
	Sentec Sensor defective	If the problem persists, contact qualified service personnel.

4.3.4 tCOM+ specific troubleshooting

Table 5 tCOM+ specific troubleshooting

Problem	Possible Cause(s)	Recommended action(s)
P0400/P0401 tCOM+ not operational, display not illuminated,	tCOM+ switched off	Switch on tCOM+.
	tCOM+ not connected to mains and battery depleted	Connect the tCOM+ with the external power supply to the mains. Allow battery to charge. If problem persists, contact qualified service personnel.
	External power supply defective	Replace tCOM+ power supply by a new one.

Problem	Possible Cause(s)	Recommended action(s)
LED bar not illuminated	tCOM+ defective	Contact your local Sentec Representative.
P0402 Fully charged Battery is discharged after less than required runtime according to section 3.2.13 “Capacity of Rechargeable Lithium Battery”	Insufficient capacity of the rechargeable Lithium Battery	Contact qualified service personnel.
	Insufficient capacity of the rechargeable Lithium Battery	Replace battery according to section 5.8 *“Replacement, Battery”*.
	tCOM+ defective	Contact your local Sentec representative.
P0403/P0504 Display is dark or not readable (speaker or fan audible or led bar/sensor illuminated)	tCOM+ in Sleep Mode	Press ON/OFF button. If the problem persists, switch the tCOM+ off and on again. If problem persists, contact qualified service personnel.
	Brightness not adjusted correctly.	Adjust brightness in Main Menu/Display
	tCOM+ defective	Contact your local Sentec representative.
P0405 POST screen does not disappear	POST failed	Switch the tCOM+ off and on again. If problem persists, contact qualified service personnel.
	tCOM+ defective	Contact your local Sentec representative.
P0406 Screen does not react if tapped or swiped	Home Mode is active	Refer to P0420
	tCOM+ Software is not responding	Switch the monitor off and on again. If it‘s not responding a switch off by long press (10s) may be required. Supervisor alarm may be released. Which will stop after proper startup. If problem persists, contact qualified service personnel.
	tCOM+ defective	Contact your local Sentec representative.
P0407 ‘Clock battery low’ displayed	Internal watch / clock battery is low	Must be replaced, contact qualified service personnel. **Note:** Refer to Status Messages ‘Clock battery low’ and ‘Set Date / Time’.
P0408 Wrong / unstable	Actual atmospheric pressure is fluctuating too much	Avoid moving the monitor in fast elevators or helicopters, or between rooms of different pressure or through airlocks.

Problem	Possible Cause(s)	Recommended action(s)
barometric pressure	Barometer chip defective	Verify that the barometric pressure reading of the tCOM+ is stable during calibration and does not deviate from a known calibrated reference barometer by more than 3 mmHg (0.4 kPa). Otherwise contact qualified service personnel. **Note:** The 'Barometer Icon' only displays if PCO_2 is enabled, and the Sentec TC Sensor is connected to the tCOM+ and inside the Docking Station. **Note:** Also refer to Status Messages 'Atm. P. unstable' and 'Barometer fault'.
	tCOM+ defective	Contact your local Sentec representative.
P0411 No audible alarms	Auditory alarm signals are PAUSED or OFF	Reactivate auditory alarm signals (if necessary)
	Volume is on minimum	Adjust the volume in the menu "Alarm Settings".
	Speaker is defective	Verify that 3 beeps are audible during POST. If problem persists, contact qualified service personnel.
	tCOM+ defective	Contact your local Sentec representative.
P0411a Buzzer is not audible at startup (1 very short, high pitched beep)	tCOM+ defective	Contact your local Sentec representative.
P0412 Battery is not charging	Connector of External power supply not entirely plugged to 12V receptacle of tCOM+	Plug in connector entirely and properly secure it. Make sure there is no gap between knurled round locking nut and hex nut of receptacle.
	External power supply defective	Replace External power supply
	tCOM+ defective	Contact your local Sentec representative.
P0414 Wrong or no analog output	Wrong Channel assignment or ranges configuration.	Check configuration of the analog output according to IFU.
	tCOM+ defective or wrong / defective cable	Contact qualified service personnel.
	tCOM+ defective	Contact your local Sentec representative.
P0415	Wrong serial communication protocol	Make sure to select appropriate serial protocol

Problem	Possible Cause(s)	Recommended action(s)
No communi-cation over RS-232	wrong / defective cable	Contact qualified service personnel.
	tCOM+ defective	Contact your local Sentec representative.
P0417 Safety test does not comply	tCOM+ defective	Contact your local Sentec representative.
P0418 Gas Level Indicator shows unexpected value	Gas Bottle empty or valve defective	Replace gas bottle by a fresh one. If problem persists, try the bottle in another tCOM+ or contact qualified service personnel.
	tCOM+ defective	Contact your local Sentec representative.
P0419 Despite a Sentec TC Sensor is connected to the tCOM+ the message 'Connect sensor' displays	Refer to P0300	
P0420 tCOM+ Main Menu cannot be accessed.	tCOM+ is in Home Mode with restricted Menu access. Tapping opens a lock screen.	The Home care organization can exit the home mode by tapping "Unlock" in the lock screen and entering the PIN which was set in the profile by the Push Power Button and select "Home Mode". To lock the screen again, press the power button, and tap "home mode".

4.3.5 Docking Station-specific troubleshooting

Table 6 Docking Station-specific troubleshooting

Problem	Possible Cause(s)	Recommended action(s)
P0500 Docking Station door does not close properly	Sensor cable jammed in the door	Adjust cable and close the Docking Station door properly.
	Sensor not inserted properly in the Docking Station door.	Insert the sensor properly and close the door.
	Mechanical defect of the Docking Station door	Contact qualified service personnel.
	Mechanical defect of the Docking Station door	Replace Docking Station door according to section 5.6 *"Replacement, Docking Station Door"*.

Problem	Possible Cause(s)	Recommended action(s)
	Gasket not placed correctly	Place gasket correctly in the chamber in the Docking Station according to section 5.4 *"Docking Station Gasket"*.
	tCOM+ defective	Contact your local Sentec representative.
P0501 Message 'Docking Station Fault' is displayed	Contaminations (e.g., gel residues) disturb the gas flow	Clean Docking Station and Sentec TC Sensor and then insert the sensor into the Docking Station. If problem persists, contact qualified service personnel. **Note:** Refer to the tCOM+ Instruction Manual.
	Docking Station defective	Contact your local Sentec representative.
P0502 Message 'Docking Station Leak: Check Gasket' is displayed	Sensor not properly positioned in Docking Station during last calibration.	Verify that the sensor is clean and is properly placed into the holder at the inside of the Docking Station door and that the door properly closes (refer to the tCOM+ Instruction Manual).
	Gasket of Docking Station contaminated	Clean the Docking Station gasket with a cotton swab with 70% isopropanol. Remove any residual threads and fibers and allow to dry. **Note:** Always clean the sensor before inserting it into the Docking Station.
	Gasket of Docking Station defective or missing	Carefully inspect the gasket and verify that it is present and well embedded in its notch. If damaged contact qualified service personnel.
	Docking Station door and / or closing mechanism defective	Carefully inspect the door and the closing mechanism for mechanical or functional damage. The closing mechanism must properly lock in place. If damaged contact qualified service personnel.
	Docking Station door mechanically defective	Replace Docking Station door according to section 5.6 *"Replacement, Docking Station Door"*.
	Gasket defective / missing	Place gasket correctly in the chamber in the Docking Station according to section 5.4 *"Docking Station Gasket"*.
	tCOM+ defective	Contact your local Sentec representative.
P0503 Calibration Gas bottle cannot be installed properly	Plastic lid of the new Calibration Gas bottle not removed	Remove the plastic lid and install the Calibration Gas bottle.
	Foreign body in the slot for the Calibration Gas bottle	Switch off the tCOM+, disconnect it from the mains supply and remove the foreign body from the slot on the rear panel of the tCOM+.
	Thread of the Calibration Gas bottle is damaged	Use a new Calibration Gas bottle and install it as described in tCOM+ Instruction Manual.
	tCOM+ defective	Contact qualified service personnel.
	tCOM+ defective	Contact your local Sentec representative.

Problem	Possible Cause(s)	Recommended action(s)
P0504 Consumption of the Calibration Gas is higher than usual (message 'Docking Station Leak: Check Gasket' does not display)	Unstable sensor requiring too many calibrations	Re-membrane sensor and observe gas consumption. If problem persists, contact qualified service personnel.
	Gasket in the Docking Station Module thread for the Calibration Gas is polluted or damaged	Replace gasket in the Docking Station Module thread according to section 5.5 *"Replacement, Gas Bottle Gasket"*.
	tCOM+ defective	Contact your local Sentec representative.
P0505 Sensor calibration does not start, Sensor LED is illuminated	Refer to P0308	
P0506 Gas Level Indicator shows wrong indication	Refer to P0418	

5 Replacement of Service Parts

To perform a safety check and for service or repair, contact qualified service personnel or your local Sentec representative. Please note that repair procedures which require opening the cover (except battery compartment) of the tCOM+ must be performed by Sentec only.

The following parts are serviceable items that may be replaced by qualified service personnel of the responsible organization with a relevant technical education/degree, e.g., medical technicians.

- Docking Station Gasket
- Docking Station Door
- Gas Bottle Gasket
- AC power adapter
- Battery
- Battery Cover
- Handle
- Cable Clip
- Silicon Foot

NOTE: Use only accessories and spare parts supplied or recommended by Sentec. Do not perform other service and repair activities than specified and described by Sentec. Failure to comply may result in physical injury, inaccurate measurements, and/or damage to the device.

Contact your local Sentec representative or Sentec service personnel if you need assistance to perform these steps.

Prior to maintenance or service all components of the SDMS must be cleaned and disinfected. If equipment is sent to service personnel, a local Sentec representative or to Sentec a completed *"Certificate for Disinfection"* (*see section* 8.1) must be attached.

If you need to return devices for service or repair, follow the shipment instructions in *section* 8.1 *"Instructions for Shipment"*.

5.1 Spare Parts

NOTE: Use only testing accessories and spare parts supplied or recommended by Sentec. Use of other than supplied or recommended parts may result in physical injury, inaccurate measurement and/or damage to the device.

Contact your local Sentec sales representative for Spare Parts and Sentec specific tools.

5.2 Required Tools

Tools

- Soft (e.g. plastic) tweezers



- Screwdriver TX10



- 2.5mm Allen/hex key (ball head optional)



- Cutter knife



- Gas Bottle Gasket replacement Toolkit (Sentec specific)



NOTE: Maintenance and repairs requiring removal of the case or covers may only be carried out by Sentec authorized service personnel trained by Sentec AG or accredited partners.
NOTE: Use only accessories and spare parts supplied or recommended by Sentec. Do not perform other service and repair activities than specified and described by Sentec. Failure to comply may result in physical injury, inaccurate measurements, and/or damage to the device.

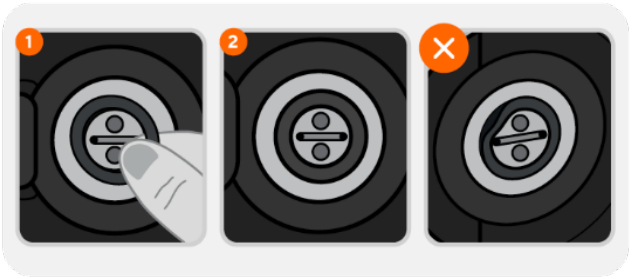
5.3 Prior to Replacement

1. If necessary, clean and disinfect the device as described in the corresponding section of the tCOM+ Instruction Manual.
2. Switch the tCOM+ off.
3. Disconnect all connections on the rear panel of the device.

5.4 Replacement, Docking Station Gasket

In case of a Docking Station leak or a possible damage, qualified service personnel, e.g., medical technicians, should replace the Docking Station Gasket.

1. Clean or disinfect hands. Open Docking Station. Remove the rubber ring (Gasket) as depicted. Alternatively, it is possible to use a pair of soft plastic tweezers.



2. Place a new Gasket onto the groove of the Docking Station, gently press around the circumference of the Gasket to push it into the groove, using thumb or finger. Note that the calibration chamber (light grey, in the center of the Docking Station) has a suspension mechanism and may therefore be pushed inwards without any risk to the monitor.

Make sure the Gasket sits tightly and evenly in the Docking Station groove and does not bulge outwards. Dispose of the used Gasket.

5.5 Replacement, Gas Bottle Gasket

It is recommended to replace the Gas Bottle Gasket annually to prevent it from baking-in (which can damage the Docking Station module), or in case of elevated calibration gas consumption, or visual damage to the gasket.

Replacement shall be done by qualified service personnel, e.g., medical technicians using the specific tool set, consisting of a removal hook and an applicator tool.

1. Remove gas bottle.
2. Inspect the gas bottle compartment visually
3. Carefully poke the removal hook into the old gasket, without scratching the brass surface, and pull out the old gasket.
4. Make sure all debris from brittle gaskets (black) are removed gasket grove (brass)
5. Load a new gasket into the applicator tool.
6. Insert the applicator tool into the gas bottle thread and push the plunger on its back end.
7. Retract the sleeve of the tool while still pushing on the plunger, to deposit the gasket.
8. Check if the gasket was properly fitted in the groove behind the thread.
9. Insert the gas bottle again and check if the thread is turning smoothly.
10. Switch on monitor, connect sensor and place sensor into docking station to initiate a PCO_2 calibration and make sure there are no docking station faults or bottle loose errors.

5.6 Replacement, Docking Station Door

To replace the Docking Station door:

1. Follow the procedure described in section 5.3 *"Prior to Replacement"*.

2. Open the Docking Station door for approx. 20mm and insert cutter knife between door and upper hinge.
3. Carefully turn the cutter knife clockwise to bend the hinge slightly upwards and disengage the door at the same time. Pay attention not to damage the hinge.
4. Insert the new Docking Station door by holding it vertically and 45° open, so the lower fin does not damage the internal pressure spring. First press on the bottom hinge to engage, then on the top one. Make sure the door is moving smoothly and closes properly.
5. Switch on the monitor.
6. Put the sensor into the Docking Station to start a calibration and a leak test.



Insert cutter knife



Disengage Docking Station door


Insert new Docking Station door

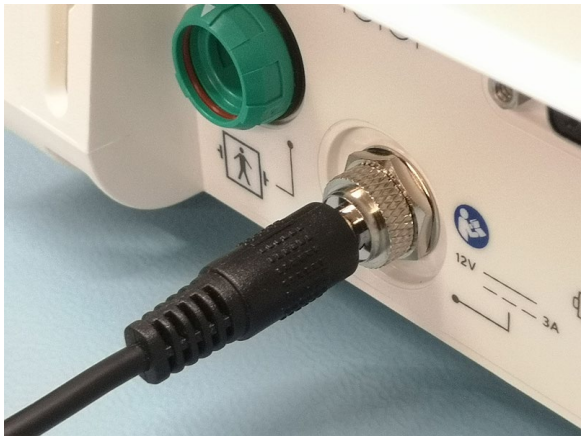
5.7 Replacement, AC Power Adapter

Use only original AC power adapter type provided by Sentec.
The power supply is available in two versions. Make sure to replace by respective version:
Hospital Use - Class II with functional earth connection (Hospital use).
Home Use - Class II without functional earth connection (Home use).

1. For this replacement the tCOM+ can stay operational.
2. Disconnect the defective power supply from mains.
3. Loosen the locking nut and disconnect the DC power plug from the back panel of tCOM+ and replace the power supply by a new one. Tighten the locking nut.
4. Check if the battery is charging.



Unlocked and disconnected



Connected and locked

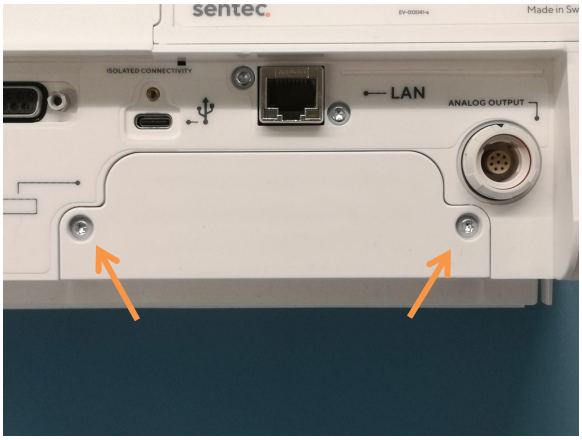
5.8 Replacement, Battery

Only use the same type of battery as replacement (RRC2057, 7.2 V / 49.7 Wh).

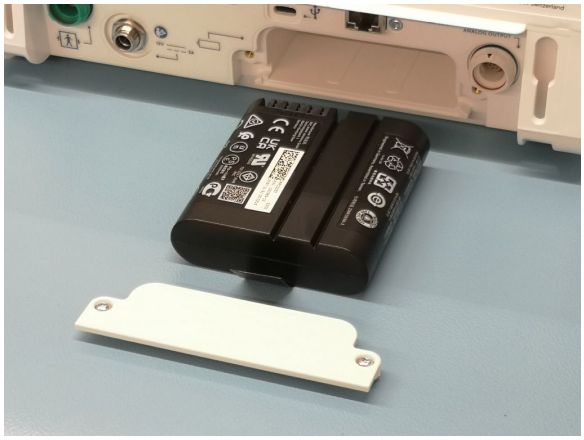
The performance and degradation of the battery highly depends on the usage scenario of the battery. Never let the battery fully discharge. Make sure to recharge it at least every 6 months.

Perform the following steps to replace the battery:

1. Open the battery cover on the back of the monitor by releasing two TX10 screws.
2. Remove the old battery by pulling on the strip.
3. Insert the new battery in the correct orientation (notches facing up and inwards).
4. Close the battery cover by means of a torque screwdriver (0.7 Nm).



Open/close battery cover


Remove/Insert battery

5.9 Replacement, Handle

1. Follow the procedure described in section 5.3 *"Prior to Replacement "*.
2. Loosen both bolts of the handle by using a 2.5mm hex key and pull out the brass sleeves.
3. Replace the defective handle by a new one.
4. Insert the brass sleeves as far as possible and tighten the bolts until flush with the handle surface (0.2 Nm).
5. Verify that:
   a. the handle can be flipped open.
   b. the handle clicks when flipping it back to the low position.
   c. the handle sticks properly to the monitor.



Brass sleeve removed



Bolt head flush with handle surface

5.10 Replacement, Cable Clip

1. Follow the procedure described in section 5.3 *"Prior to Replacement"* .
2. Remove the defective cable clip by lifting and pulling out.
3. Insert the new cable clip into the slot and lock by pushing down.



Removing Cable clip

5.11 Replacement, Silicon Foot

If the Silicon Foot got lost or extremely contaminated, perform the following steps to replace the Silicon Foot:

1. Follow the procedure described in section 5.3 *"Prior to Replacement "*.
2. Turn the monitor upside down. Make sure the LED bar does not get damaged by putting it on a soft cushion.
3. Remove damaged silicon feet.
4. Clean the groove for the silicon feet with a cotton swap soaked with 70% isopropanol.
5. Insert the new silicon feet into the groove by pressing firmly.

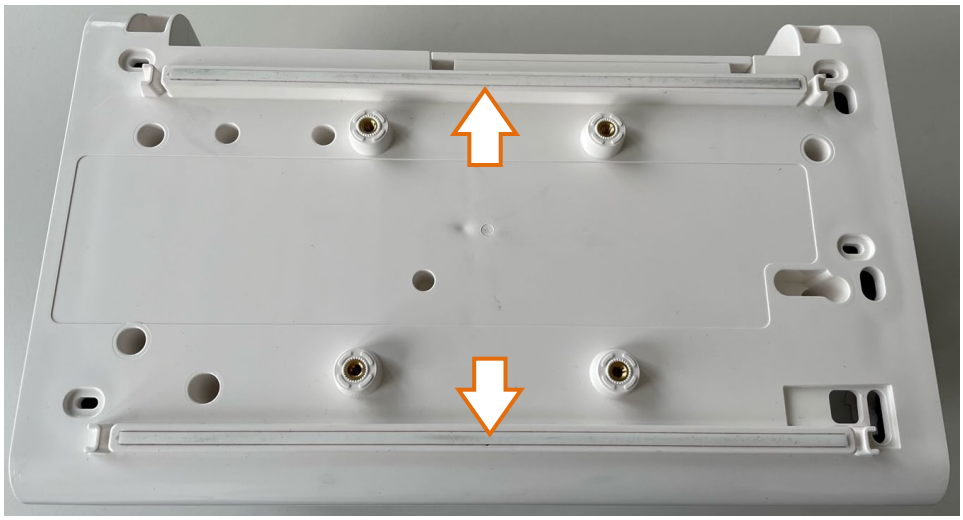


Figure 13 Replacement, Silicon Foot

6 Software Updates

The software of the tCOM+ can be upgraded via the DATA/SERVICE USB Port on the left side of the monitor. This procedure is to be carried out by qualified service personnel with a relevant technical education/degree, e.g., medical technicians.

1. Connect a USB type-C flash drive with the corresponding software to the port
2. Open the Menu and go to Advanced settings. Enter the admin password that was set during the initial setup of the monitor.
3. Choose ‘Software Update'. This will show the currently installed software version and the available version on the USB flash drive.
4. Tap ‘Start update’ to automatically start the upgrade of the relevant components. The process is fully automated and takes approx. 15 minutes.

Ensure to check the software version for accuracy in the System Information menu once the software update is completed.

Note: Software updates must be reported to Sentec by using the Software Update Reporting Form to ensure traceability.

7 Disposal

The SDMS contains electronic printed circuit boards, a display, cables, and lithium batteries.

Do not incinerate equipment or gas bottles.

WEEE Disposal: European consumers are obliged by law to dispose Waste Electrical and Electronic Equipment (WEEE) according to the WEEE Directive:

1. All electrical and electronic waste, must be stored, collected, treated, recycled, and disposed of separately from other waste
2. Consumers are obliged by law to return electrical and electronic devices at the end of their service lives to the public collection points set up for this purpose or

point of sale. Details to this are defined by the national law of the respective country.

7.1 Patient Monitor

Return the tCOM+ to your local Sentec representative or dispose it according to local regulations.

7.2 Cables

Dispose the cables according to local regulations. The copper contained can be recycled.

7.3 Sentec TC Sensors

Return the Sentec TC Sensors to your local distributor.

7.4 Calibration Gas Bottle

Dispose the gas bottles according to local regulations for lined aluminum containers. Make sure that only empty gas bottles are disposed.

Gas may be discharged from the container by **carefully** opening the container valve. Ensure that the container is positively supported. Open the container valve **slowly** to permit gas discharge at an appropriate rate.

NOTE: Ensure that the operation is carried out in a well-ventilated area and vented gases may disperse. Noise level should be controlled to meet local regulations.

⚠ **WARNING:** Pressurized container. Protect from sunlight and do not expose to temperatures exceeding 50°C (122°F). Do not pierce or burn, even after use. Do not spray on a naked flame or any incandescent material.

7.5 Consumables

All material used is considered "non-critical". The consumables may be disposed with the regular garbage collection.

8 Appendix

8.1 Forms

Note: The forms listed below can be found on www.sentec.com/ifu.

8.1.1 Protocol for Safety & Functionality Test

This protocol is used as checklist and to document the results of the tests.

HBQ-229

8.1.2 Customer Feedback Form

Can be used for all kinds of feedback, e.g., software update of a single device.

HBQ-48 (HB-006872)

8.1.3 Software Update Reporting Form

To be used to report software updates.

MMQ-83 (RF-012921)

8.1.4 Certificate of Disinfection

Please refer to the document:

HBQ-6 (HB-005637)

8.1.5 SDMS Maintenance Checklist

To be used to schedule and keep track of cleaning/maintenance by the operator.

HBQ-59 (HB-007601)

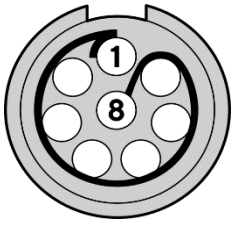
8.1.6 Device Service Request

Country-/region-specific forms for submitting any warranty claims on Sentec products.

FO-159

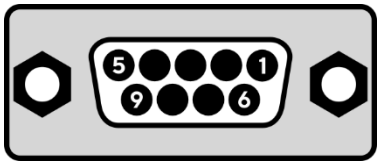
8.2 Interfaces - Pin Assignment

8.2.1 Sensor Connection Port



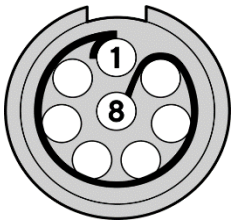
Pin(s)	Signal
1	Supply Ground
2, 8	Signal Ground
3	(reserved)
4	Received Data (RX, from Sensor)
5	5 V, for sensor supply only
6	Transmitted Data (TX, to Sensor)
7	3.3 V, for sensor supply only

8.2.2 Serial Data Port (RS-232)



Pin(s)	Signal
1	(reserved)
2	Transmitted Data (TX, from tCOM+)
3	Received Data (RX, to tCOM+)
4	(reserved)
5	Signal Ground
6 – 9	(reserved)

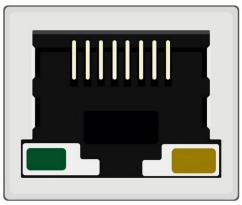
8.2.3 Analog Output Port



Pin(s)	Channel	Signal (default)
1	CH 0	Pleth
7	CH 1	PR
4	CH 2	SpO_2
8	CH 3	PCO_2
2	Signal Ground	
3	(reserved)	
5	(reserved)	
6	(reserved)	

Note: Signal configuration can be changed in the Interfaces menu.

8.2.4 Network Port (LAN)



The Network Port of the tCOM+ is used to communicate with external computer-based data collection systems. The interface allows remote monitoring and download of historical trend data. The Network Port is located on the rear panel of the tCOM+. The Network Port is a standard RJ45 Ethernet connector.

8.2.5 Data/Service USB Port

The Data/Service USB Port is located on the left side panel of the tCOM+ monitor. The USB Port can be used for connecting USB-Sticks for software updates, import and export of profiles or measurement data.

8.2.6 Isolated Connectivity USB Port

The Isolated Connectivity Port of the tCOM+ may be used to connect other devices in the future. It is currently not in use. The Isolated Connectivity Port is located on the rear panel of the tCOM+.

8.3 Technical Specifications

See the corresponding section of the tCOM+ instruction manual.

8.4 Contact

Phone: +41 61 726 97 60

Monday to Friday, 08:30 - 12.00 und 14.00 - 17:00 (GMT +01:00)

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